Sin Nombre hantavirus interacts with helminth parasite co-infections in Montana deer mice (*Peromyscus maniculatus*)

Madeline Rowland[1], Mike Kinsella[2], Amy Kuenzi[3], and Angela Luis[1]

[1]*W.A. Franke College of Forestry and Conservation, University of Montana, Missoula, MT*

[2]*Helm West Labs, Missoula, MT*

[3]*Department of Biological Sciences, Montana Technological University, Butte, MT*

madeline.rowland@umontana.edu

Introduction

- **Wildlife** are often **co-infected** with multiple **parasite** species simultaneously
- Wild **deer mice** are the **reservoir host for Sin Nombre hantavirus** (**SNV**), which causes **Hantavirus Pulmonary Syndrome** in humans with a **case fatality rate of ~ 35%**[1]
- Deer mice are also infected with **helminth** parasites[2]
- ***Trichuris stansburyi*** is a soil-transmitted helminth that causes mild gastrointestinal pathology
- ***Calodium hepaticum*** requires the death of the host for transmission and causes severe liver pathology

★ We ask:

(1) if SNV interacts with helminths in the host
(2) how host size influences infection status and
(3) how host sex influences co-infections

Methods

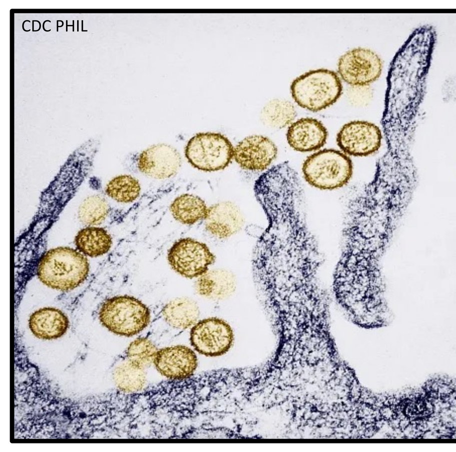
- **Small mammal trapping** at six 1-ha Sherman trap grids from May to November 2023 - 2025 in **western Montana**
- Deer mice euthanized in the field and dissected in the lab to **identify *Trichuris* and *Calodium* infections**
- Blood samples tested for **SNV antibodies**[5]
- **Multispecies distribution models** compared with NIMBLE to determine (1) whether infections interact and the effects of (2) host size and (3) sex on infection status[6]

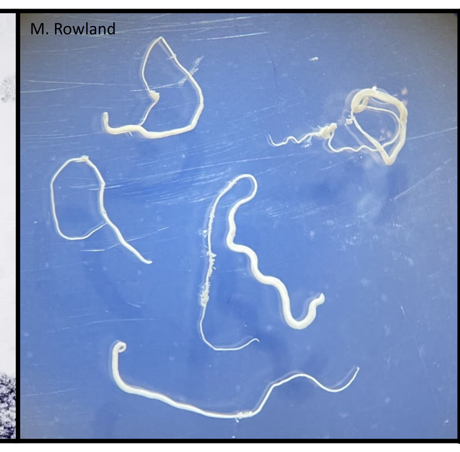
Sherman trap with grid location marker

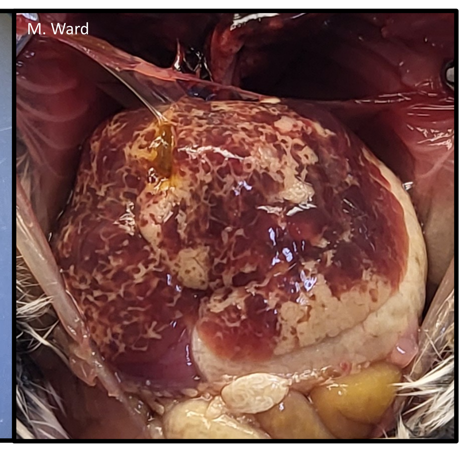
M. Rowland collects samples in the field

Preliminary Results

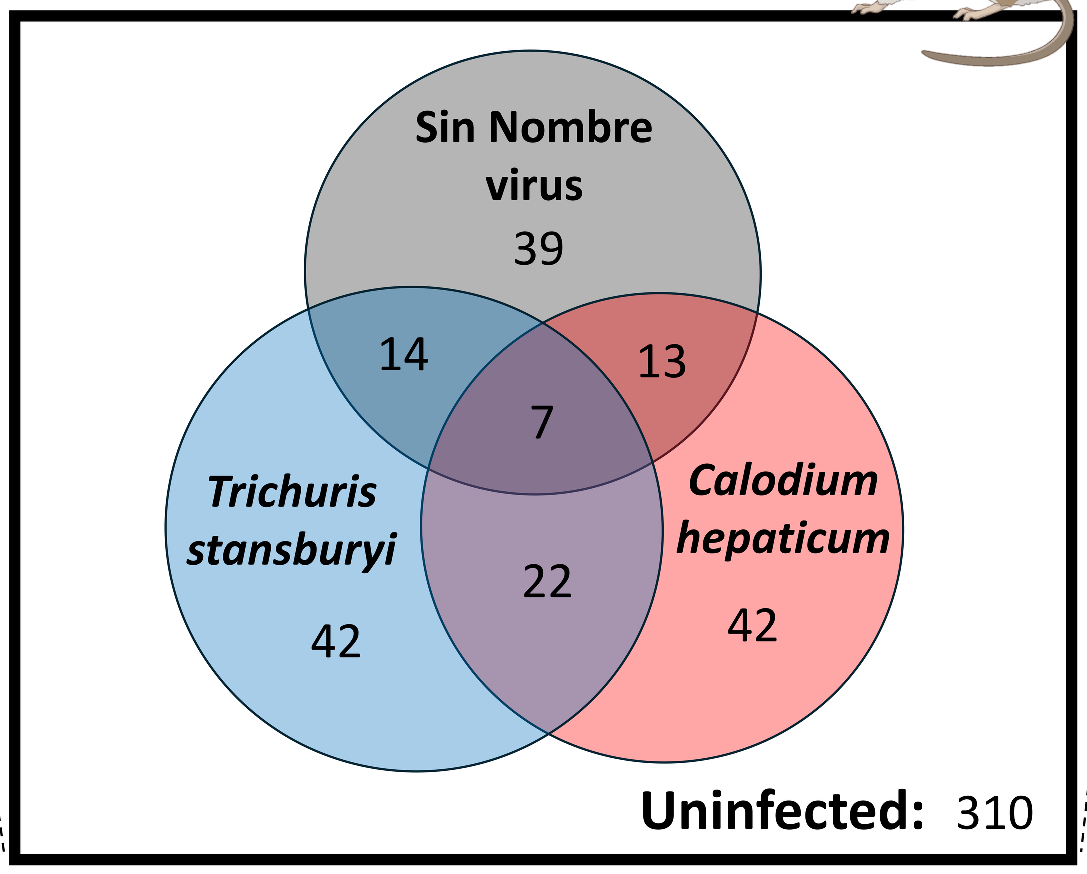
Sin Nombre virus (SNV)	*Trichuris stansburyi*	*Calodium hepaticum*



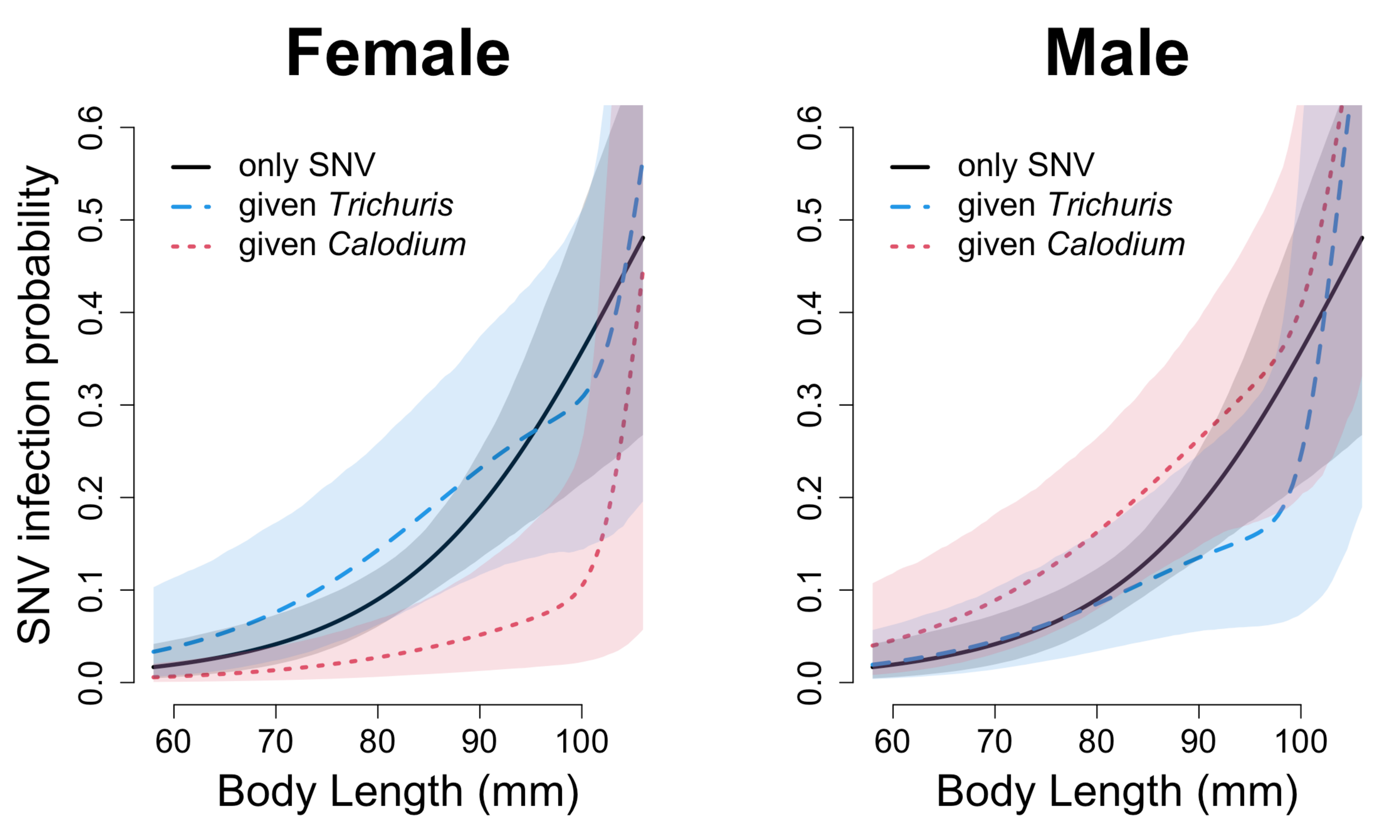




Total: 489* of 1,065 deer mice



***Dissections are in progress**. These data include all SNV+ mice across three years and only 42% of SNV- mice from the total sample (n=1,065). As a result, SNV infection is inflated here (15%) compared to the true sample proportion (6.9%).



Model predicted SNV infection probability with 95% CRIs for female and male deer mice as a function of body length for SNV alone (black), conditional on *T. stansburyi* infection (blue) and conditional on *C. hepaticum* infection (red). SNV infection probability was significantly lower in females and higher in males co-infected with *C. hepaticum.*

Conclusions

- Model predicted **SNV infection probability** was **lower for females** and **higher for males that were infected with *Calodium hepaticum***
- **We found strong support for this sex-specific interaction between SNV and *Calodium hepaticum***
- In contrast, we found marginal support for a sex-specific interaction between SNV and *Trichuris stansburyi* in the opposite direction
- As expected, we found strong support for an effect of **body length on all single infections** likely because bigger, older individuals had a longer exposure period

★ The top model included:

(1) parasite interactions
(2) an effect of size on single infections, and
(3) an effect of sex on interactions

Future Directions

- **Increase sample size** (576 dissections remaining)
- Incorporate **other infections** (Cestodes, Protozoans, Acanthocephalans, other Nematodes, Arthropods, etc.) and **infection intensities** in the model framework
- Incorporate **mechanisms of parasite interaction** including host immune health and body condition
- Compare the immune response and identify **immune tradeoffs** among deer mice with different combinations of parasite infections by **quantifying immune gene expression** with **RNA sequencing**[7]

Acknowledgments

We thank the Confederated Salish and Kootenai Tribes, MPG Ranch, the City of Missoula, Spirit Stables LLC, and the Harpers for access to their properties. Thank you to the University of Montana and Montana Tech students who assisted with data collection, especially C. Karnatz, R. Poindexter, M. Asanovich, F. Cruse, A. Calonico, D. Kelly, P. Todd, and R. Throckmorton. Financial support was provided by the NSF EEID grant 2109828. The animation was created with BioRender.

[1]Parmenter et al. 1993, [2]Ubelaker and Rácz 2022, [3]Frandsen and Grundmann 1961, [4]Fuehrer 2013, [5]Kuenzi et al. 2005, [6]Fidino et al. 2019, [7]Britton et al. 2023